

Intrinsic vs. Post-hoc Explainability in Credit Risk Prediction:

A Quantitative Evaluation of Faithfulness, Stability, and Predictive Performance

Abubakar Imran Waqas Ahmed Muhammad Huzaifa Khalid
i232589@isb.nu.edu.pk i232540@isb.nu.edu.pk i232508@isb.nu.edu.pk

Department of Artificial Intelligence & Data Science
School of Computing, FAST-NUCES

Abstract—Credit-risk prediction is a Crucial tabular classification problem in which predictive accuracy alone is not sufficient for deployment, because lending decisions also require calibrated probabilities, operationally meaningful thresholds, and defensible explanations. This paper compares intrinsic and post-hoc explainability strategies for credit default prediction on the UCI Default of Credit Card Clients dataset. The evaluated models include Logistic Regression, LightGBM, CatBoost, a multilayer perceptron (MLP) with mixed tabular inputs, and TabNet. Explanations are produced with TreeSHAP for gradient-boosted trees, Integrated Gradients for the MLP, and TabNet attention masks for the intrinsic model. Using a repeated 5-fold, 3-seed protocol over the engineered-feature pipeline, we jointly evaluate discrimination, calibration, perturbation-based faithfulness, explanation stability, runtime, cost-sensitive thresholding, and descriptive group-wise diagnostics.

The results show that no single model-explainer pair dominates every criterion. CatBoost achieves the strongest predictive performance among the core uncalibrated models, with mean ROC-AUC 0.7890 ± 0.0095 and PR-AUC 0.5666 ± 0.0150 . MLP with Integrated Gradients produces the highest faithfulness scores, including Faithfulness@5 of 0.2377, while LightGBM with TreeSHAP yields the highest stability, with mean Spearman correlation 0.8683. TabNet masks are the fastest explanations on a per-sample basis at approximately 0.036 ms/sample, but they are substantially less stable than the tree- and gradient-based alternatives. These findings suggest that explanation reliability depends on the model-explainer pair rather than on whether a method is labeled intrinsic or post-hoc, and that deployment choices should be aligned with predictive, interpretive, and operational objectives jointly.

Index Terms—Credit risk prediction, tabular data, SHAP, TabNet, explainable AI, intrinsic interpretability, post-hoc interpretability, Integrated Gradients, XGBoost, LightGBM, MLP

I. INTRODUCTION

Credit-risk prediction is a key decision-support task in consumer finance, affecting loan approval, credit limits, pricing, and portfolio risk directly. In these situations, classification errors have asymmetric consequences: false negatives can expose lenders to default losses, while false positives can reject creditworthy applicants and reduce business opportunity. Thus a useful lending model must provide not only strong

discrimination but also calibrated risk estimates, threshold selection and intelligible explanations.

Linear and tree-based methods have dominated the tabular credit datasets in the past and still hold their own thanks to their good performance on heterogeneous financial variables. Meanwhile, deep learning tabular architectures have been proposed to capture non-linear feature interactions in a more flexible manner. The interpretability problem is then even more pronounced: more powerful function classes may lead to better predictions in certain aspects, but they are typically more difficult to justify without additional explanation techniques.

This has motivated two broad explainability directions. Post-hoc methods, such as SHAP [1] and Integrated Gradients [2], explain already-trained models. Intrinsic approaches attempt to build interpretability into the architecture itself, with TabNet being a widely discussed example in tabular learning and credit scoring [3]. However, the presence of an intrinsic explanation mechanism should not be treated as evidence that the resulting explanations are automatically faithful, stable, or suitable for deployment.

Many applied studies in credit-risk modelling emphasize predictive metrics and qualitative explanation plots. Fewer studies jointly evaluate predictive performance, calibration, explanation faithfulness, explanation stability, explanation cost, and cost-sensitive operating thresholds under a practical compute budget. This paper addresses that gap by benchmarking transparent, boosted-tree, deep, and intrinsically explainable tabular models within one repeated cross-validation pipeline.

This paper makes four contributions:

- 1) We provide a repeated cross-validation benchmark of intrinsic and post-hoc explainability methods for credit-risk prediction on mixed tabular data.
- 2) We compare Logistic Regression, LightGBM, CatBoost, MLP, and TabNet using predictive metrics, calibration, explanation faithfulness, stability, runtime, and cost-sensitive evaluation.
- 3) We evaluate explanation quality quantitatively using perturbation-based faithfulness, ranking stability, ran-

dom explanation baselines, and runtime analysis rather than relying only on visual explanations.

- 4) We show that explanation quality varies by model-explainer pair and cannot be inferred from the intrinsic/post-hoc label alone.

II. RELATED WORK

A. Credit-risk prediction

Credit-risk prediction has long served as a benchmark for tabular supervised learning. The UCI Default of Credit Card Clients dataset remains a standard reference point for evaluating the probability of default [4]. Classical methods, like Logistic Regression, continue to be appealing because of their interpretability and efficiency, while boosted decision trees are often more accurate for financial data. Deep learning approaches have also been studied with class imbalance correction and feature extraction in recent years. [5].

B. Deep tabular learning

Tabular deep learning design remains a challenging task in itself because tabular problems benefit from thoughtful choices of inductive bias rather than simply stacking layers. From broad surveys, it can be observed that tree ensemble-based models are not easy to beat in practice in the realm of tabular datasets [6]. With regard to credit risk prediction, TabNet architectures were considered a promising approach to combining nonlinearity with in-built feature selection capabilities [3]. The MLP architecture has proven to remain a useful baseline due to its capacity to capture complicated relationships and support gradient-based attribution techniques.

C. Explainable AI for tabular data

Explanations for post-hoc explainability of tabular models are primarily driven by feature-attribution techniques like SHAP [1] and Integrated Gradients [2]. The usage of SHAP and its tree-based variations is common practice in credit risk research since they can be utilized with powerful tree-based models while supporting both the global and local perspectives of explanations. [5]. Integrated gradients provide a coherent attribution technique for differentiable models, making it appropriate for multilayer perceptron networks. Intrinsic approaches, such as TabNet masks, offer another interesting option since they are computed during inference time.

However, visual plausibility is not enough. Explanations based on saliency maps can be found to lack even basic robustness criteria [7], and explanation evaluation requires explicit metrics such as infidelity, sensitivity, perturbation tests, and ranking stability [8], [9]. Previous research has focused on either maximizing prediction accuracy or providing visualizations; in our research, we quantitatively evaluate explanations with computational limits in mind.

III. DATASET AND PROBLEM FORMULATION

The experiments are performed on the UCI Default of Credit Card Clients dataset [4] which is a binary classification problem where positive class is default in the

next month. The authoritative experimental setup is the extended run with engineered features as described in `results/config.json`. Duplicate rows were removed before modelling, reducing the sample count from 30,000 to 29,965. The final experimental matrix contains 38 predictors: 29 numerical variables and 9 categorical or ordinal-coded variables. The observed default rate in the cleaned dataset is 22.1%.

The modelling pipeline uses the original UCI predictors together with domain-inspired engineered credit-risk features. These engineered variables summarize recent billing and repayment behaviour through utilization ratios, payment-to-bill ratios, bill and payment aggregates, recent change indicators, and delayed-payment summaries. This distinction is important because the main reported benchmark does not operate on the raw 23 predictors alone.

Table I summarizes the verified dataset properties used in the paper.

Let $\mathbf{x} \in \mathbb{R}^d$ denote the feature vector for a client and let $y \in \{0, 1\}$ denote the default label, where $y = 1$ indicates default next month. Each model estimates a probability $\hat{p}(y = 1 | \mathbf{x})$. Performance is measured on held-out outer test folds, while threshold-dependent deployment analyses use thresholds selected on inner validation splits only.

IV. METHODOLOGY

A. Preprocessing

The preprocessing pipeline removes the identifier field and drops duplicate rows before splitting. Category cleaning follows the notebook implementation: EDUCATION values 0, 5, and 6 are remapped to 4, while MARRIAGE value 0 is remapped to 3. Numerical scaling and categorical encoding are fit on training data within each fold and then applied to the corresponding validation and test partitions. No statistics from the outer test fold are used during preprocessing, calibration, or threshold selection. This design is intended to prevent test leakage.

B. Feature engineering

The engineered feature set augments the original repayment history, bill amounts, and payment amounts with domain-motivated summaries. These include current and average utilization, average and maximum payment delay, the number of delayed months, average and maximum bill and payment amounts, their dispersion, payment-to-bill ratios, and recent changes in bills and payments. The raw-versus-engineered ablation is reported separately to distinguish representation effects from model effects.

C. Models

Five model families are compared:

- **Logistic Regression:** a transparent linear baseline with strong interpretability and low computational cost.
- **LightGBM:** a gradient-boosted tree ensemble that is typically strong on mixed tabular features and efficient to train.

TABLE I
DATASET SUMMARY FOR THE AUTHORITATIVE ENGINEERED-FEATURE EXPERIMENT.

Property	Value
Original samples	30,000
Samples after duplicate removal	29,965
Duplicates removed	35
Target	Default payment next month
Positive class rate	22.1%
Number of original predictors	23
Number of final features	38
Numerical features	29
Categorical/ordinal features	9
Cross-validation setting	5 folds $\times$ 3 seeds

- **CatBoost**: a boosted-tree method included because it is a strong tabular baseline and performed best in the final benchmark.
- **MLP**: a feed-forward neural network for tabular classification, used here as a black-box deep baseline compatible with gradient attribution.
- **TabNet**: an intrinsic deep tabular architecture that produces attention-like feature masks during inference [3].

D. Explanation methods

The explanation methods are aligned to the particular model family:

- **TreeSHAP** for CatBoost and LightGBM [1].
- **Integrated Gradients** for the MLP [2].
- **TabNet masks** as the intrinsic explanation channel for TabNet.
- **Model-specific random baselines** to assess if observed faithfulness and stability were above chance-level rankings.

E. Explanation evaluation metrics

Faithfulness is evaluated with a top- k masking protocol over $k \in \{1, 3, 5, 10\}$. For each sample, the top-ranked features are perturbed and the resulting change in predicted probability is measured. Larger mean probability change indicates that the explainer identified features with greater local predictive influence. As with other perturbation-based tests, these results should be interpreted cautiously because feature masking can create unrealistic samples [8].

Stability is measured as the Spearman rank correlation of global feature rankings across seeds and folds. Higher correlation indicates that the explanation method produces more reproducible importance rankings under repeated data resampling and retraining.

Runtime is measured for both model training and explanation generation. Explanation runtime is reported per sample to reflect deployment-relevant cost.

Calibration is assessed with the Brier score and expected calibration error (ECE). Additional calibrated MLP variants are included because calibration is important when decision thresholds are cost-sensitive.

Cost-sensitive evaluation uses three lender scenarios with different false-negative and false-positive costs. Thresholds are

selected on validation data and then applied unchanged to the outer test fold.

Group-wise diagnostics report descriptive differences in default rate, predicted positive rate, recall, false-positive rate, false-negative rate, Brier score, and ROC-AUC across selected demographic variables. These diagnostics are auditing aids rather than causal fairness claims.

V. EXPERIMENTAL SETUP

The final paper uses only the verified extended engineered-feature configuration. The repeated evaluation protocol uses 5 outer folds and 3 random seeds, giving 15 runs per model. The pipeline considers 500 explanation samples per run, 100 background samples for TreeSHAP, and the Integrated Gradients step settings from `results/config.json`. The saved configuration is used to enable early stopping on the neural models.

The hardware profile captured for the final run shows four CPU cores and two Tesla T4 GPUs, with CUDA enabled. This hardware suffices for the benchmark but still renders explanation runtime materially relevant, in particular for repeated cross-validation and deployment-oriented auditing.

Table II summarizes the main experimental choices.

VI. RESULTS

A. Predictive performance

Table III provides the main performance metrics. In terms of ROC-AUC, PR-AUC, F1-score, recall, balanced accuracy, and Matthews correlation coefficient, CatBoost obtains the highest average scores among the top performing algorithms without calibration. However, its superiority compared to LightGBM and Multi-layer perceptron is relatively small in line with the challenges to establish clear differentiation between highly ranked tabular baselines in this particular scenario. LightGBM performs quite well and demonstrates the best Brier score without calibration.

ROC and Precision-Recall charts are illustrated in Fig. ??, whereas the pooled confusion matrices are shown in Fig. 2. The charts support the fact that there is no absolute winner and that the problem has multiple conflicting objectives.

TABLE II
EXPERIMENTAL SETUP USED FOR THE FINAL BENCHMARK.

Item	Setting
Run mode	extended
Feature mode	engineered
Cross-validation	5-fold stratified CV
Seeds	42, 2024, 7
Runs per model	15
Models	Logistic Regression, CatBoost, LightGBM, MLP, TabNet
Explainers	TreeSHAP, Integrated Gradients, TabNet masks
Primary predictive metrics	ROC-AUC, PR-AUC, F1, recall, MCC, Brier
Explanation metrics	Faithfulness@k, stability, runtime
Calibration metrics	Brier score, ECE
Explanation samples	500 per run
SHAP background samples	100
IG steps	25 (config), 32 (n_steps)
Hardware	4 CPU cores, 2 × Tesla T4, CUDA enabled

TABLE III
PREDICTIVE PERFORMANCE OVER 15 RUNS (5 FOLDS × 3 SEEDS).

Model	ROC-AUC	PR-AUC	F1	Precision	Recall	Bal. Acc.	MCC	Brier
Logistic Regression	0.7741 ± 0.0076	0.5407 ± 0.0124	0.5367 ± 0.0120	0.5123 ± 0.0268	0.5661 ± 0.0310	0.7058 ± 0.0084	0.3980 ± 0.0171	0.1838 ± 0.0021
CatBoost	0.7890 ± 0.0095	0.5666 ± 0.0150	0.5481 ± 0.0129	0.5235 ± 0.0331	0.5788 ± 0.0337	0.7136 ± 0.0087	0.4131 ± 0.0197	0.1770 ± 0.0028
LightGBM	0.7825 ± 0.0104	0.5555 ± 0.0147	0.5421 ± 0.0123	0.5254 ± 0.0321	0.5639 ± 0.0390	0.7086 ± 0.0100	0.4076 ± 0.0180	0.1537 ± 0.0013
MLP	0.7828 ± 0.0088	0.5563 ± 0.0146	0.5405 ± 0.0118	0.5281 ± 0.0282	0.5567 ± 0.0347	0.7069 ± 0.0092	0.4067 ± 0.0155	0.1809 ± 0.0025
TabNet	0.7731 ± 0.0099	0.5350 ± 0.0135	0.5311 ± 0.0143	0.5156 ± 0.0319	0.5528 ± 0.0462	0.7014 ± 0.0114	0.3938 ± 0.0170	0.1846 ± 0.0051

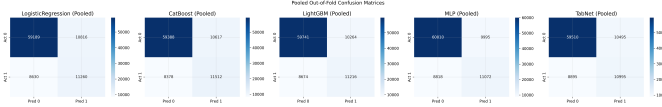


Fig. 2. Pooled confusion matrices across the repeated evaluation protocol.

B. Statistical testing

The results of the Wilcoxon signed-rank test related to the performance of the CatBoost model are displayed in Table IV. For the measures of ROC-AUC, PR-AUC, and F1, CatBoost performs better compared to the rest of the competing models for each metric, with the maximum mean difference observed against TabNet and Logistic Regression. For the CatBoost vs. LightGBM comparison, there is a relatively smaller difference, but it remains preferable for CatBoost. It is worth noting that the results obtained from cross-validation are not completely independent; therefore, the reported p-values should be interpreted as supportive rather than definitive evidence.

C. Calibration

The output of the calibration process is shown in Table V and Figure 3 below. In the set of models not yet calibrated, LightGBM is seen to be superior to all other models since it records the best Brier score and ECE, suggesting its probabilities have the best quality. However, when calibrated, the MLP model shows remarkable improvement in both criteria: MLP with Platt scaling achieves the best Brier score overall, while MLP with isotonic calibration achieves the lowest ECE.

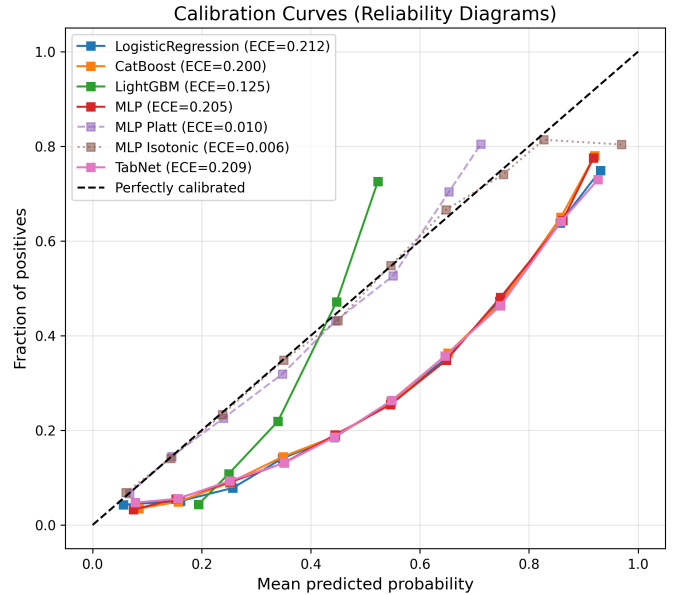


Fig. 3. Calibration curves for the evaluated models.

D. Explanation faithfulness

Table VI contains faithfulness results for top- k masking, whereas Figure 4 depicts the faithfulness plot along with the random baselines. The MLP network paired with Integrated Gradients shows the best faithfulness scores at any given k value, including Faithfulness@1 of 0.1151 and Faithfulness@10 of 0.2651. CatBoost with TreeSHAP is the second-best approach, with TabNet masks in third place and the LightGBM network with TreeSHAP in fourth place. Notably, all actual explainers perform better than the random baselines specific to each model. This suggests that the measured

TABLE IV
CATBOOST-FOCUSED STATISTICAL COMPARISONS FROM REPEATED-FOLD SUMMARIES.

Comparison	Metric	Mean diff.	Median diff.	p -value	Direction
CatBoost vs LightGBM	ROC-AUC	0.0065	0.0071	0.000061	CatBoost > LightGBM
CatBoost vs MLP	ROC-AUC	0.0063	0.0065	0.000122	CatBoost > MLP
CatBoost vs TabNet	ROC-AUC	0.0159	0.0165	0.000061	CatBoost > TabNet
CatBoost vs Logistic Regression	ROC-AUC	0.0150	0.0155	0.000061	CatBoost > Logistic Regression
CatBoost vs LightGBM	PR-AUC	0.0111	0.0124	0.000122	CatBoost > LightGBM
CatBoost vs MLP	PR-AUC	0.0103	0.0119	0.000183	CatBoost > MLP
CatBoost vs TabNet	PR-AUC	0.0316	0.0295	0.000061	CatBoost > TabNet
CatBoost vs Logistic Regression	PR-AUC	0.0259	0.0259	0.000061	CatBoost > Logistic Regression
CatBoost vs LightGBM	F1	0.0060	0.0078	0.002014	CatBoost > LightGBM
CatBoost vs MLP	F1	0.0076	0.0077	0.000122	CatBoost > MLP
CatBoost vs TabNet	F1	0.0170	0.0161	0.000061	CatBoost > TabNet
CatBoost vs Logistic Regression	F1	0.0114	0.0117	0.000061	CatBoost > Logistic Regression

TABLE V
CALIBRATION METRICS. LOWER IS BETTER FOR BOTH Brier SCORE AND ECE.

Model	Brier	ECE
Logistic Regression	0.1838	0.2121
CatBoost	0.1770	0.2000
LightGBM	0.1537	0.1250
MLP	0.1809	0.2048
MLP (Platt)	0.1347	0.0104
MLP (Isotonic)	0.1348	0.0059
TabNet	0.1846	0.2092

rankings capture nontrivial predictive influence, although the absolute values remain contingent on the perturbation design.

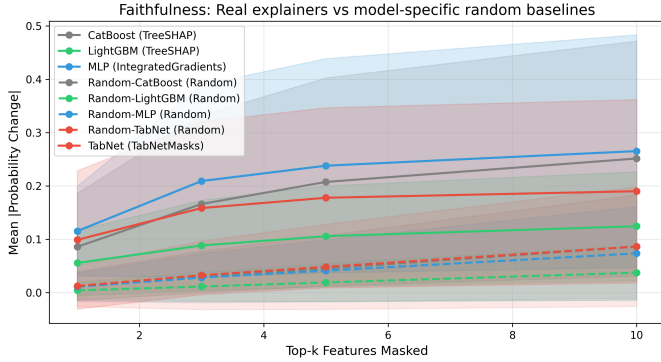


Fig. 4. Faithfulness curves for real explainers and model-specific random baselines.

E. Explanation stability

Table VII and Fig. 5 summarize ranking stability across seeds and folds. LightGBM with TreeSHAP is the most stable method with mean Spearman $\rho = 0.8683$, followed closely by CatBoost with TreeSHAP at $\rho = 0.8584$. MLP with Integrated Gradients is also stable, but less so than the tree-based explainers. TabNet masks have the weakest stability by a substantial margin, with mean $\rho = 0.2394$.

This result matters because stability reflects reproducibility: if global importance rankings vary substantially under retraining, then the explanation channel becomes difficult to rely on for audit or governance workflows.

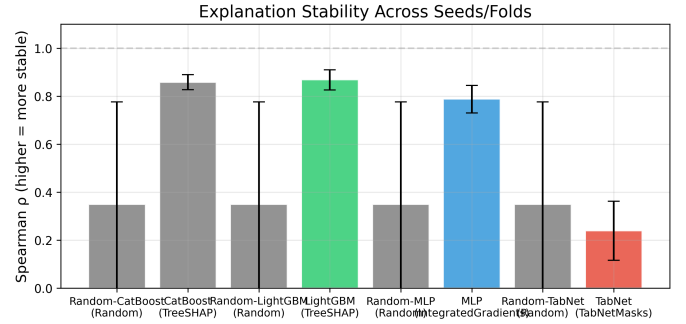


Fig. 5. Explanation stability across seeds and folds.

F. Runtime

Table VIII and Fig. 6 report runtime. LightGBM is by far the fastest model to train, with mean training time of 0.21 seconds per run. CatBoost and the MLP have similar training costs near 10 seconds, while TabNet is the slowest by a wide margin at 30.85 seconds on average.

For explanation generation, TabNet masks are the fastest at approximately 0.036 ms/sample, followed closely by LightGBM TreeSHAP at 0.046 ms/sample. CatBoost TreeSHAP and MLP Integrated Gradients are both substantially slower, each above 0.6 ms/sample. Thus, the fastest explanation channel in this benchmark is not the most faithful or the most stable one.

TABLE VI
PERTURBATION-BASED FAITHFULNESS. HIGHER MEAN PROBABILITY CHANGE INDICATES STRONGER MEASURED FAITHFULNESS.

Model	Explainer	Faithfulness@1	Faithfulness@3	Faithfulness@5	Faithfulness@10
CatBoost	TreeSHAP	0.0856	0.1659	0.2072	0.2512
LightGBM	TreeSHAP	0.0553	0.0881	0.1056	0.1242
MLP	Integrated Gradients	0.1151	0.2089	0.2377	0.2651
TabNet	TabNet masks	0.0987	0.1583	0.1776	0.1899
Random-CatBoost	Random	0.0112	0.0292	0.0448	0.0862
Random-LightGBM	Random	0.0038	0.0111	0.0188	0.0371
Random-MLP	Random	0.0109	0.0278	0.0409	0.0733
Random-TabNet	Random	0.0123	0.0323	0.0481	0.0861

TABLE VII
EXPLANATION STABILITY ACROSS REPEATED RUNS. HIGHER SPEARMAN ρ INDICATES MORE STABLE RANKINGS.

Model	Explainer	Mean Spearman ρ
CatBoost	TreeSHAP	0.8584 $\pm$ 0.0310
LightGBM	TreeSHAP	0.8683 $\pm$ 0.0418
MLP	Integrated Gradients	0.7877 $\pm$ 0.0571
TabNet	TabNet masks	0.2394 $\pm$ 0.1234
Random-CatBoost	Random	0.3494 $\pm$ 0.4273
Random-LightGBM	Random	0.3494 $\pm$ 0.4273
Random-MLP	Random	0.3494 $\pm$ 0.4273
Random-TabNet	Random	0.3494 $\pm$ 0.4273

TABLE VIII
TRAINING AND EXPLANATION RUNTIME.

Model	Explainer	Training time (s)	Explanation time (s)	Explanation time/sample (ms)
Logistic Regression	not reported	1.6875 $\pm$ 0.1943	not reported	not reported
CatBoost	TreeSHAP	10.7002 $\pm$ 2.3094	0.3609 $\pm$ 0.0796	0.7219
LightGBM	TreeSHAP	0.2135 $\pm$ 0.0211	0.0232 $\pm$ 0.0031	0.0465
MLP	Integrated Gradients	9.9105 $\pm$ 2.7384	0.3068 $\pm$ 0.0141	0.6136
TabNet	TabNet masks	30.8522 $\pm$ 7.3617	0.0182 $\pm$ 0.0009	0.0364

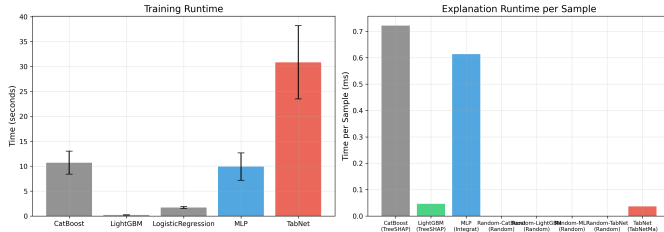


Fig. 6. Training-time and explanation-time comparison.

G. Cost-sensitive evaluation

Table IX reports the best model under three lender scenarios. In all three cases, CatBoost yields the lowest test average cost after validation-based threshold selection. The selected thresholds vary meaningfully by scenario, which illustrates why credit-risk deployment cannot be reduced to a single fixed decision threshold.

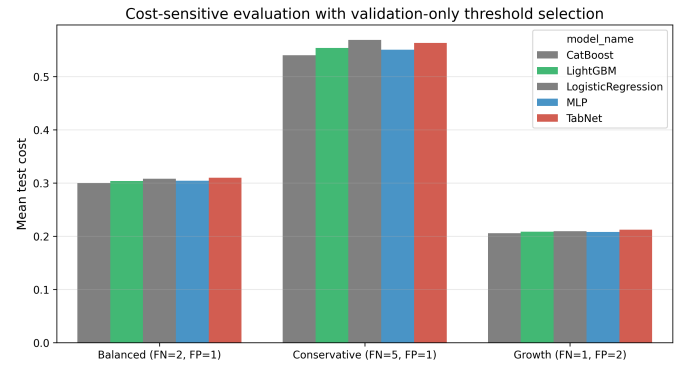


Fig. 7. Validation-selected thresholds under alternative lender cost assumptions.

H. Trade-off summary

Table X consolidates the main trade-offs. CatBoost offers the strongest overall predictive and cost-sensitive profile. LightGBM provides the best balance of training efficiency, uncalibrated calibration, and explanation stability. The MLP yields the strongest perturbation-based faithfulness with Integrated Gradients. TabNet provides very fast intrinsic explana-

TABLE IX
COST-SENSITIVE DEPLOYMENT SUMMARY USING VALIDATION-SELECTED THRESHOLDS.

Scenario	Best model	Threshold	Avg. cost	(FN, FP)
Balanced	CatBoost	0.6200	0.3000	(2, 1)
Conservative	CatBoost	0.3807	0.5402	(5, 1)
Growth	CatBoost	0.8753	0.2059	(1, 2)

tions, but not the strongest predictive or stability profile in this setting.

I. Fairness diagnostics and local case studies

The group-wise results indicate that error rates and predicted positive rates vary across SEX, EDUCATION, and MARRIAGE groups. These results should only be considered descriptive diagnoses, not causative fairness or unfairness diagnoses. To support the global analyses, we have included four local case studies, which include examples of true positive, true negative, false positive, and false negative classifications. The examples prove that there can be different high-ranked features for similar output predictions even among different explainers.

VII. DISCUSSION

Three main trends can be identified from the results.

First, it becomes clear that CatBoost shows better results in terms of predictive performance compared to LightGBM and MLP; however, the gap between them is not significant. This information is valuable since in practice, discrimination measures alone cannot be the only factors determining the use of one model vs another, as some other differences can be observed as well.

Second, the model which produces the most consistent and stable explanations turns out to be a different one from the one producing explanations that best correspond to the predictions. Thus, the MLP trained with Integrated Gradients shows the best faithfulness measure in terms of perturbation-based approaches, whereas LightGBM shows higher stability using TreeSHAP. Both of these characteristics are valuable under different circumstances – while faithfulness matters in case local attribution is needed, stability becomes important in case of auditing and policy reviews.

Third, the results do not support the assumption that intrinsic interpretability automatically produces more reliable explanations. Although TabNet supplies built-in feature masks and achieves the fastest per-sample explanation runtime, its stability is markedly weaker than TreeSHAP and Integrated Gradients in this benchmark. In other words, explanation quality must be measured empirically rather than inferred from architecture labels alone.

From a deployment perspective, the implications are straightforward. If the primary objective is strongest discrimination and lowest cost under the evaluated lender scenarios, CatBoost is the most defensible choice. If the goal is stable and relatively well-calibrated uncalibrated probability estimation with very fast training and explanation, LightGBM is attractive. If the goal is strongest perturbation-based faithfulness, the MLP paired with Integrated Gradients is more compelling.

Explanations in credit-risk systems should therefore be selected in tandem with the predictive model and the intended operational objective, not bolted on as an afterthought.

VIII. THREATS TO VALIDITY AND LIMITATIONS

However, there are many restrictions that have been observed in this research study. First, the benchmark depends solely on the usage of a publicly available dataset, and the UCI credit card default dataset does not necessarily capture the present-day credit approval processes, population dynamics, or organizational factors. Second, although using multiple cross-validations improves the accuracy of the variance calculation, this technique does not produce completely independent data sets; hence, statistical tests must be treated carefully. Third, the faithfulness measure is highly dependent on the masking process.

Fourth, engineered features are domain-inspired and may not transfer directly without further validation in other lending settings. Fifth, TabNet may respond to additional tuning beyond the saved configuration, so the reported results should be interpreted as evidence for this benchmark setting rather than as a universal verdict on the architecture. Sixth, explanation methods rest on different assumptions, which complicates direct comparison across model families. Finally, the group-wise diagnostics are descriptive only, and the cost-sensitive scenarios are simplified operational abstractions rather than complete institutional decision models.

IX. CONCLUSION

This paper evaluated intrinsic and post-hoc explainability strategies for credit-risk prediction using predictive, calibration, explanation-quality, runtime, and deployment-oriented metrics. The results show that no single model-explainer pair dominates every criterion. CatBoost delivers the strongest predictive and cost-sensitive performance in the evaluated setting, MLP with Integrated Gradients produces the highest perturbation-based faithfulness, LightGBM with TreeSHAP produces the strongest stability and best uncalibrated calibration, and TabNet provides the fastest per-sample explanations without matching the strongest stability or predictive outcomes.

The central conclusion is therefore practical rather than ideological: credit-risk XAI systems should be selected according to deployment priorities, because explanation quality depends on the model-explainer pair and cannot be inferred from whether a method is intrinsic or post-hoc.

TABLE X
HIGH-LEVEL TRADE-OFF SUMMARY FOR THE MAIN BENCHMARK.

Model	ROC-AUC	Faith@5	Stability	Train (s)
Logistic Regression	0.7741	not reported	not reported	1.7
CatBoost	0.7890	0.2072	0.8584	10.7
LightGBM	0.7825	0.1056	0.8683	0.2
MLP	0.7828	0.2377	0.7877	9.9
TabNet	0.7731	0.1776	0.2394	30.9

APPENDIX A ADDITIONAL RESULTS

A. Raw vs. engineered feature ablation

Table XI reports the raw-versus-engineered ablation. Engineered features improve ROC-AUC for LightGBM, CatBoost, and MLP, but not every downstream metric improves uniformly. In particular, engineered features improve Brier score for all three compared models, while F1 is slightly lower for LightGBM and MLP than in the corresponding raw-setting summary. This reinforces that feature engineering changes both ranking quality and operating-threshold behavior.

B. All-pairs statistical summary

Table XII expands the statistical analysis beyond the CatBoost-focused view used in the main text.

C. Descriptive group-wise diagnostics

Tables XIII, XIV, and XV summarize the provided group-wise audit outputs. These tables are descriptive diagnostics only and should not be interpreted as causal fairness analysis.

D. Local case studies

Table XVI reports four canonical prediction cases. The goal is not to assert causal explanations, but to show how the leading features identified by CatBoost TreeSHAP, MLP Integrated Gradients, and TabNet masks differ across the same outcome category.

TABLE XI
RAW-VERSUS-ENGINEERED ABLATION SUMMARY FROM THE PROVIDED RESULTS.

Feature setting	Model	ROC-AUC	PR-AUC	F1	Brier
Raw	LightGBM	0.7840	0.5596	0.5456	0.1339
Raw	CatBoost	0.7857	0.5647	0.5450	0.1333
Raw	MLP	0.7831	0.5545	0.5388	0.1792
Engineered	LightGBM	0.7904	0.5614	0.5442	0.1327
Engineered	CatBoost	0.7951	0.5686	0.5506	0.1319
Engineered	MLP	0.7859	0.5542	0.5330	0.1772

TABLE XII
ALL-PAIRS STATISTICAL COMPARISONS FROM THE REPEATED BENCHMARK.

Comparison	Metric	Mean diff.	Median diff.	p-value	Direction
LightGBM vs MLP	ROC-AUC	-0.0003	-0.0001	0.977966	MLP > LightGBM
LightGBM vs TabNet	ROC-AUC	0.0094	0.0093	0.000061	LightGBM > TabNet
MLP vs TabNet	ROC-AUC	0.0096	0.0097	0.000061	MLP > TabNet
CatBoost vs LightGBM	ROC-AUC	0.0065	0.0071	0.000061	CatBoost > LightGBM
CatBoost vs MLP	ROC-AUC	0.0063	0.0065	0.000122	CatBoost > MLP
CatBoost vs TabNet	ROC-AUC	0.0159	0.0165	0.000061	CatBoost > TabNet
CatBoost vs Logistic Regression	ROC-AUC	0.0150	0.0155	0.000061	CatBoost > Logistic Regression
LightGBM vs MLP	PR-AUC	-0.0009	0.0002	0.934082	MLP > LightGBM
LightGBM vs TabNet	PR-AUC	0.0205	0.0159	0.000061	LightGBM > TabNet
MLP vs TabNet	PR-AUC	0.0213	0.0161	0.000061	MLP > TabNet
CatBoost vs LightGBM	PR-AUC	0.0111	0.0124	0.000122	CatBoost > LightGBM
CatBoost vs MLP	PR-AUC	0.0103	0.0119	0.000183	CatBoost > MLP
CatBoost vs TabNet	PR-AUC	0.0316	0.0295	0.000061	CatBoost > TabNet
CatBoost vs Logistic Regression	PR-AUC	0.0259	0.0259	0.000061	CatBoost > Logistic Regression
LightGBM vs MLP	F1	0.0016	0.0007	0.524475	LightGBM > MLP
LightGBM vs TabNet	F1	0.0110	0.0092	0.000305	LightGBM > TabNet
MLP vs TabNet	F1	0.0094	0.0072	0.000122	MLP > TabNet
CatBoost vs LightGBM	F1	0.0060	0.0078	0.002014	CatBoost > LightGBM
CatBoost vs MLP	F1	0.0076	0.0077	0.000122	CatBoost > MLP
CatBoost vs TabNet	F1	0.0170	0.0161	0.000061	CatBoost > TabNet
CatBoost vs Logistic Regression	F1	0.0114	0.0117	0.000061	CatBoost > Logistic Regression

TABLE XIII
GROUP-WISE DIAGNOSTICS FOR SEX.

Model	Group	Size	Default rate	Pred. pos. rate	Recall	FPR	FNR	Brier
Logistic Regression	SEX=1	35622	0.2416	0.2813	0.5946	0.1814	0.4054	0.1953
Logistic Regression	SEX=2	54273	0.2079	0.2222	0.5444	0.1376	0.4556	0.1762
CatBoost	SEX=1	35622	0.2416	0.2779	0.6008	0.1751	0.3992	0.1865
CatBoost	SEX=2	54273	0.2079	0.2253	0.5620	0.1370	0.4380	0.1707
LightGBM	SEX=1	35622	0.2416	0.2614	0.5758	0.1612	0.4242	0.1609
LightGBM	SEX=2	54273	0.2079	0.2242	0.5548	0.1375	0.4452	0.1489
MLP	SEX=1	35622	0.2416	0.2727	0.5889	0.1719	0.4111	0.1936
MLP	SEX=2	54273	0.2079	0.2092	0.5320	0.1245	0.4680	0.1725
TabNet	SEX=1	35622	0.2416	0.2710	0.5780	0.1731	0.4220	0.1947
TabNet	SEX=2	54273	0.2079	0.2181	0.5335	0.1353	0.4665	0.1780

TABLE XIV
GROUP-WISE DIAGNOSTICS FOR EDUCATION.

Model	Group	Size	Default rate	Pred. pos. rate	Recall	FPR	FNR	Brier
Logistic Regression	EDUCATION=1	31689	0.1924	0.1957	0.5144	0.1198	0.4856	0.1668
Logistic Regression	EDUCATION=2	42057	0.2374	0.2698	0.5839	0.1721	0.4161	0.1928
Logistic Regression	EDUCATION=3	14745	0.2517	0.3011	0.6125	0.1964	0.3875	0.2034
Logistic Regression	EDUCATION=4	1404	0.0705	0.0605	0.2121	0.0490	0.7879	0.0918
CatBoost	EDUCATION=1	31689	0.1924	0.1994	0.5376	0.1189	0.4624	0.1595
CatBoost	EDUCATION=2	42057	0.2374	0.2671	0.5890	0.1669	0.4110	0.1860
CatBoost	EDUCATION=3	14745	0.2517	0.3059	0.6300	0.1968	0.3700	0.1969
CatBoost	EDUCATION=4	1404	0.0705	0.0477	0.1616	0.0391	0.8384	0.0949
LightGBM	EDUCATION=1	31689	0.1924	0.1948	0.5226	0.1168	0.4774	0.1424
LightGBM	EDUCATION=2	42057	0.2374	0.2573	0.5751	0.1583	0.4249	0.1593
LightGBM	EDUCATION=3	14745	0.2517	0.2935	0.6106	0.1869	0.3894	0.1659
LightGBM	EDUCATION=4	1404	0.0705	0.1125	0.2222	0.1042	0.7778	0.1100
MLP	EDUCATION=1	31689	0.1924	0.1788	0.5005	0.1021	0.4995	0.1613
MLP	EDUCATION=2	42057	0.2374	0.2607	0.5776	0.1621	0.4224	0.1905
MLP	EDUCATION=3	14745	0.2517	0.2940	0.6017	0.1905	0.3983	0.2008
MLP	EDUCATION=4	1404	0.0705	0.0726	0.2121	0.0621	0.7879	0.1250
TabNet	EDUCATION=1	31689	0.1924	0.1922	0.5008	0.1187	0.4992	0.1674
TabNet	EDUCATION=2	42057	0.2374	0.2607	0.5735	0.1634	0.4265	0.1920
TabNet	EDUCATION=3	14745	0.2517	0.2918	0.5912	0.1911	0.4088	0.2033
TabNet	EDUCATION=4	1404	0.0705	0.0933	0.2222	0.0835	0.7778	0.1580

TABLE XV
GROUP-WISE DIAGNOSTICS FOR MARRIAGE.

Model	Group	Size	Default rate	Pred. pos. rate	Recall	FPR	FNR	Brier
Logistic Regression	MARRIAGE=1	40929	0.2346	0.2651	0.5803	0.1685	0.4197	0.1908
Logistic Regression	MARRIAGE=2	47835	0.2095	0.2280	0.5527	0.1420	0.4473	0.1772
Logistic Regression	MARRIAGE=3	1131	0.2361	0.2812	0.5581	0.1956	0.4419	0.2080
CatBoost	MARRIAGE=1	40929	0.2346	0.2603	0.5890	0.1595	0.4110	0.1810
CatBoost	MARRIAGE=2	47835	0.2095	0.2334	0.5701	0.1442	0.4299	0.1729
CatBoost	MARRIAGE=3	1131	0.2361	0.2759	0.5393	0.1944	0.4607	0.2076
LightGBM	MARRIAGE=1	40929	0.2346	0.2448	0.5615	0.1478	0.4385	0.1573
LightGBM	MARRIAGE=2	47835	0.2095	0.2329	0.5666	0.1445	0.4334	0.1502
LightGBM	MARRIAGE=3	1131	0.2361	0.2803	0.5506	0.1968	0.4494	0.1681
MLP	MARRIAGE=1	40929	0.2346	0.2517	0.5719	0.1535	0.4281	0.1870
MLP	MARRIAGE=2	47835	0.2095	0.2185	0.5427	0.1326	0.4573	0.1750
MLP	MARRIAGE=3	1131	0.2361	0.2750	0.5318	0.1956	0.4682	0.2093
TabNet	MARRIAGE=1	40929	0.2346	0.2504	0.5583	0.1560	0.4417	0.1880
TabNet	MARRIAGE=2	47835	0.2095	0.2285	0.5479	0.1439	0.4521	0.1811
TabNet	MARRIAGE=3	1131	0.2361	0.2750	0.5393	0.1933	0.4607	0.2139

TABLE XVI
LOCAL EXPLANATION CASE STUDIES.

Case	True	Pred.	Prob.	CatBoost SHAP top features	MLP IG top features	TabNet mask top features	Short interpretation
TP	1	1	0.8978	PAY_0; num_delayed_months; avg_pay_delay; max_pay_delay; LIMIT_BAL	PAY_0; num_delayed_months; max_pay_delay; LIMIT_BAL; PAY_2	max_pay_delay; PAY_2; num_delayed_months; avg_pay_amt; PAY_AMT2	All three methods emphasize delinquency history and persistent repayment stress.
TN	0	0	0.2949	utilization_current; PAY_0; avg_utilization; max_pay_delay; num_delayed_months	avg_utilization; utilization_current; BILL_AMT5; MARRIAGE; BILL_AMT4	utilization_current; max_pay_delay; BILL_AMT6; LIMIT_BAL; BILL_AMT2	The negative prediction is associated with lower utilization pressure and milder delinquency signals.
FP	0	1	0.5512	max_pay_delay; PAY_0; num_delayed_months; avg_pay_delay; avg_payment_to_bill_ratio	max_pay_delay; num_delayed_months; PAY_0; EDUCATION; PAY_2	max_pay_delay; avg_pay_amt; num_delayed_months; recent_payment_change; LIMIT_BAL	Delay-related attributes dominate even though the borrower ultimately does not default.
FN	1	0	0.2809	PAY_AMT1; num_delayed_months; max_pay_delay; utilization_current; payment_to_bill_ratio_1	recent_bill_change; recent_payment_change; PAY_AMT1; avg_pay_amt; std_bill_amt	recent_bill_change; max_pay_delay; BILL_AMT1; PAY_0; BILL_AMT3	Strong recent payment signals appear to suppress the positive prediction in a true default case.

REFERENCES

- [1] S. M. Lundberg and S. Lee, “A unified approach to interpreting model predictions,” in *Advances in Neural Information Processing Systems*, 2017, pp. 4765–4774.
- [2] M. Sundararajan, A. Taly, and Q. Yan, “Axiomatic attribution for deep networks,” in *Proceedings of the 34th International Conference on Machine Learning*, 2017, pp. 3319–3328.
- [3] S. Wang and X. Zhang, “Research on credit default prediction model based on tabnet-stacking,” *Entropy*, vol. 26, no. 10, p. 861, 2024.
- [4] I. Yeh and C. Lien, “The comparisons of data mining techniques for the predictive accuracy of probability of default of credit card clients,” *Expert Systems with Applications*, vol. 36, no. 2, pp. 2473–2480, 2009.
- [5] I. Aruleba and Y. Sun, “Enhanced credit risk prediction using deep learning and SMOTE-ENN resampling,” *Machine Learning with Applications*, vol. 21, p. 100692, 2025.
- [6] V. Borisov, T. Leemann, K. Seßler, J. Haug, M. Pawelczyk, and G. Kasneci, “Deep neural networks and tabular data: A survey,” *IEEE Transactions on Neural Networks and Learning Systems*, 2022, referenced via the project proposal and literature review sheet.
- [7] J. Adebayo, J. Gilmer, M. Muelly, I. Goodfellow, M. Hardt, and B. Kim, “Sanity checks for saliency maps,” in *Advances in Neural Information Processing Systems*, 2018.
- [8] C. Yeh, C. Hsieh, A. S. Suggala, D. I. Inouye, and P. Ravikumar, “On the (in)fidelity and sensitivity of explanations,” in *Advances in Neural Information Processing Systems*, 2019.
- [9] H. O. Quinn, M. Sedky, J. Francis, and M. Streeton, “Literature review of explainable tabular data analysis,” *Electronics*, vol. 13, no. 19, p. 3806, 2024.